AIROCEAN GASES

Reliable high-purity gases for semiconductor aerospace, research, and advanced manufacturing

AIROCEAN GASES

HIGH-PURITY SPECIALTY GAS SOLUTIONS



AIROCEAN GASES LLC

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AIROCEAM GASES

COMMITMENT TO EXCELLENCE IN EVERY SERVICE

POWERING ADVANCED INDUSTRIES WITH HIGH-PURITY GASES

AIROCEAN GASES ▶ OVERVIEW

Airocean Gases®, a leading chemical manufacturing company, specializes in the supply of regional specialty gases.

As one of the largest suppliers in the area, we are committed to delivering exceptional service, ensuring a reliable supply chain, and maintaining high standards of product quality。

COMMITMENT TO EXCELLENCE IN EVERY SERVICE



we provide comprehensive industrial gas solutions from basic services to advanced technologies to meet your diverse needs.

Our team brings together experienced professionals from both technical and international trade backgrounds, dedicated to delivering exceptional service and solutions to our customers.



WENTAO GUO

Master in Northeastern University
Bachelor in Iowa State University



WENHAO GUO

University of Maryland,College Park
Bachelor in University of Delaware



JIAHAO GUO

Master in New York University
Bachelor in Arizona State University

2024 — COMPANY ESTABLISHED

Focused on delivering high-purity rare gases to advanced U.S. industries.

2025 — U.S. WAREHOUSE LAUNCHED

Enabling stable local inventory and faster delivery.

2025 — JOINED GAWDA

Strengthened commitment to safety, compliance, and industry standards.

2026+ — EXPANSION & QUALITY ENHANCEMENT

Growing product lines and upgrading analytical and QA systems.

COMPANY VALUE ▼

CUSTOMER CENTRIC

Airocean Gases is dedicated to providing excellent customer service, collaborating with customers to improve productivity and finding solutions that help their businesses succeed.

ACTING FOR THE FUTURE

Airocean Gases acts for a sustainable future and builds meaningful relationships in our workplaces and with the communities and customers we serve.

COMMITTED TO SAFETY

Airocean Gases top priority and core value is the safety of our associates, customers, and the communities in which we operate.

DIVERSITY & INLCUSION

Equality, Diversity, Inclusion, Better together, everywhere we go.

AIROCEAN GASES UNDERSTANDS THE CHALLENGES OUR CUSTOMERS FACE.

Whether our customers create, build, care, serve or sustain, Airocean Gases comprehensive portfolio of products and services — backed by a network of specialists with expertise in specialty gases, bulk gases, welding, healthcare,safety and on-site supply chain management — helps customers overcome these challenges and more, every day.

POWER UTILITIES & MATERIALS

Remain competitive in domestic markets

Implement productivity and efficiency enhancements



INDUSTRIAL GAS DISTRIBUTOR

Provide reliable supply assurance for regional distributors

Consistent purity and packaging standards



AIROCEAN GASES

SERVING DIVERSE INDUSTRIES

- CONTROL COSTS, IMPROVE EFFICIENCIES AND CONSOLIDATE SUPPLIERS
- PRODUCE HIGH QUALITY PRODUCTS AND INCREASE PRODUCTION WITH MINIMAL OR NO DOWNTIME
- IMPROVE SAFETY IN THE WORKPLACE
- HELP OUR CUSTOMERS REDUCE EMISSIONS AND ACHIEVE ECO-FRIENDLY GROWTH

SEMICONDUCTOR AND ELECTRONICS

Etching and chamber cleaning

Advanced manufacturing technology

High-purity process gas

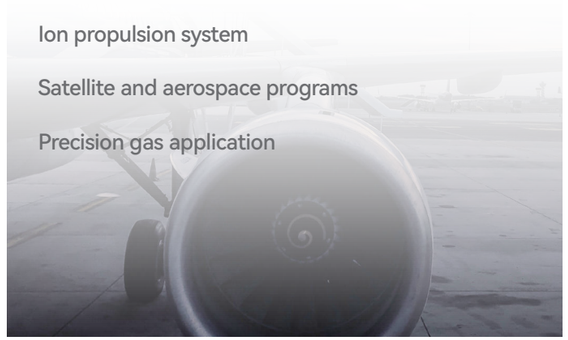


AEROSPACE AND SPACE PROPULSION

Ion propulsion system

Satellite and aerospace programs

Precision gas application



ELECTRIC POWER COMPANIES AND ELECTRICAL EQUIPMENT

GIS/GIL insulation material

High-voltage equipment testing

Electrical safety application (SF_6)



RESEARCH AND ADVANCED LABORATORY

Universities and national laboratories

Physics, optics, and materials research

Analysis and experiment system



U.S. WAREHOUSING & NATIONWIDE DISTRIBUTION

Strategically located inventory to support fast and reliable delivery across the United States.

- Local inventory stocked in Pennsylvania
- Short lead times and flexible delivery options
- Compliant storage and distribution for specialty gases



Precision Molecules for Advanced Industries

PRODUCT GUIDE

SPECIALTY GASES

RELIABLE SOURCING

Global network ensures supply stability.

QUALITY ASSURANCE

Guaranteed purity levels meeting international standards.

COMPETITIVE PRICING

Optimized logistics for cost efficiency.

Airocean Gases is a trusted global provider of high-purity specialty gases critical to advanced manufacturing, electronics, and lighting sectors. Our portfolio includes rare atmospheric gases and sophisticated compounds engineered for precision performance. We ensure stable supply chains, stringent quality control, and competitive pricing for B2B procurement professionals worldwide.

This guide provides a detailed look into seven of our core offerings:

KRYPTON **XENON** **NEON** **HELIUM** **N_2O** **SF_6** **CF_4** **WF_6**

Each gas meets the exacting demands of modern industrial processes,supported by Airocean's commitment to reliability and customer success



The Lighting & Insulation Innovator

Krypton (Kr)

Krypton (Kr) is a colorless, odorless, and non-toxic rare noble gas extracted from the air. Its unique properties, including low thermal conductivity and high atomic weight, make it indispensable for high- performance applications in lighting and specialized insulation.

CORE APPLICATIONS & AIROCEAN ADVANTAGE

HIGH-EFFICIENCY LIGHTING

Used as a filler gas in incandescent and fluorescent lamps to reduce filament evaporation and increase bulb lifespan and brightness.

THERMAL INSULATION

Ideal for insulating double-pane windows due to its superior thermal efficiency compared to argon,leading to better energy conservation.

EXCIMER LASERS

A critical component in Krypton Fluoride (KrF) excimer lasers used in semiconductor lithography.

KEY PHYSICAL PROPERTIES

Kr

AIROCEAN'S KRYPTON

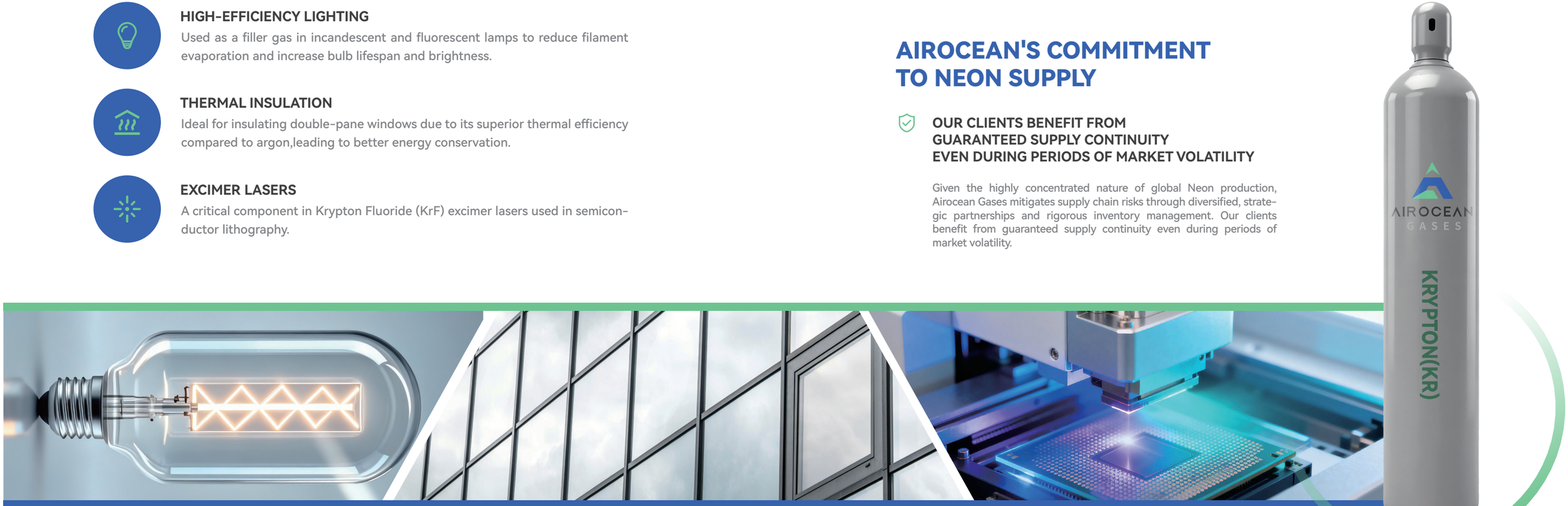
Crucial for maximizing the lifespan and efficiency of sensitive lighting and electronic components.

Chemical Formula	Kr	**Purity**	99.999%

AIROCEAN'S COMMITMENT TO NEON SUPPLY

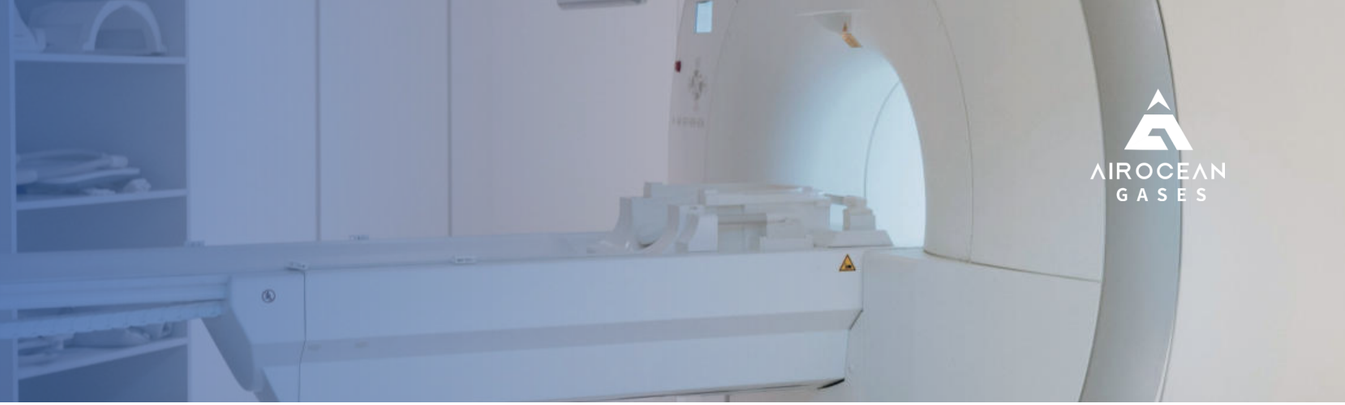
OUR CLIENTS BENEFIT FROM GUARANTEED SUPPLY CONTINUITY EVEN DURING PERIODS OF MARKET VOLATILITY

Given the highly concentrated nature of global Neon production, Airocean Gases mitigates supply chain risks through diversified, strategic partnerships and rigorous inventory management. Our clients benefit from guaranteed supply continuity even during periods of market volatility.



The High-Performance Noble Gas

Xenon (Xe)



Xenon (Xe) is the rarest and heaviest non-radioactive noble gas. Known for its intense light emission when electrically excited and its non-reactive nature, Xenon is vital for cutting-edge technologies in propulsion,

KEY INDUSTRIAL APPLICATIONS

AUTOMOTIVE LIGHTING

Essential component in high-intensity discharge (HID) lamps for superior brightness and spectral quality.

SEMICONDUCTOR ETCHING

Used in advanced etching processes in the manufacturing of microprocessors and memory chips.

PROJECTION SYSTEM

Xenon short-arc lamps power cinema projectors and specialized industrial lighting systems.

PHYSICAL SPECIFICATIONS

Xe

AIROCEAN'S XENON

Airocean Gases maintains a robust sourcing strategy for Xenon, ensuring continuity of supply despite its rarity and increasing global demand, particularly from the space industry.

Chemical Formula	Xe	**Purity**	99.9995%

OTHER APPVLICATIONS

SPACE PROPULSION

Used as propellant in ion thrusters for satellites and spacecraft due to its high atomic mass and ease of ionization.

MEDICAL IMAGING

Hyperpolarized Xenon is utilized in advanced MRI for detailed lung imaging.







The Optoelectronic Workhorse

Neon (Ne)

Neon (Ne) is a chemically inert, light noble gas known for the characteristic brilliant reddish-orange glow it emits in vacuum tubes. Beyond signage, Neon's high ionization potential and use in gas lasers make it fundamental to the electronics and optoelectronics industries.

CORE APPLICATIONS

SEMICONDUCTOR MANUFACTURING

Used as a carrier and dilution gas in plasma-based semiconductor processes, supporting process stability and impurity control.

PRECISION LASERS & METROLOGY

A key component in helium-neon (He-Ne) laser systems for precision alignment, measurement, and calibration.

RESEARCH & ADVANCED LABORATORIES

Applied in optical experiments and laboratory environments requiring consistent purity and controlled gas quality.

PERFORMANCE CHARACTERISTICS

Ne

AIROCEAN'S NEON

Ensuring reliable performance in critical lighting, signage, and high-precision electronic applications.

Chemical Formula	Ne	**Purity Range**	99.9995% to 99.9999%

AIROCEAN'S COMMITMENT TO NEON SUPPLY

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The Invisible Lifeline of High-Tech Industry

Helium (He)

Helium is the second lightest element in the universe and one of the most strategically critical gases in modern industry. As a non-renewable, non-reactive noble gas, it is irreplaceable in processes demanding extreme purity, ultra-low temperatures, or inert atmospheres — from semiconductor fabs to MRI machines.

CORE APPLICATIONS

SEMICONDUCTOR MANUFACTURING

Used as carrier and purge gas in CVD/ALD deposition processes. Helium's high thermal conductivity stabilizes plasma and prevents contamination at advanced nodes.

CRYOGENICS & MRI SYSTEMS

Liquid helium is the only substance cold enough to maintain superconducting magnets in MRI machines and particle accelerators at near absolute zero (−269°C).

FIBER OPTICS PRODUCTION

Helium atmosphere during fiber drawing prevents oxidation and surface defects, ensuring consistent optical transmission quality.

LEAK DETECTION

Standard tracer gas for vacuum system integrity testing — its small atomic size allows detection of the smallest leaks in aerospace, automotive, and industrial systems.

PERFORMANCE CHARACTERISTICS

He

AIROCEAN'S HELIUM

Precision-sourced, ultra-high purity helium for the most demanding industrial and scientific applications.

Chemical Formula	He	**Purity**	99.9999%

CYLINDER SPECIFICATIONS

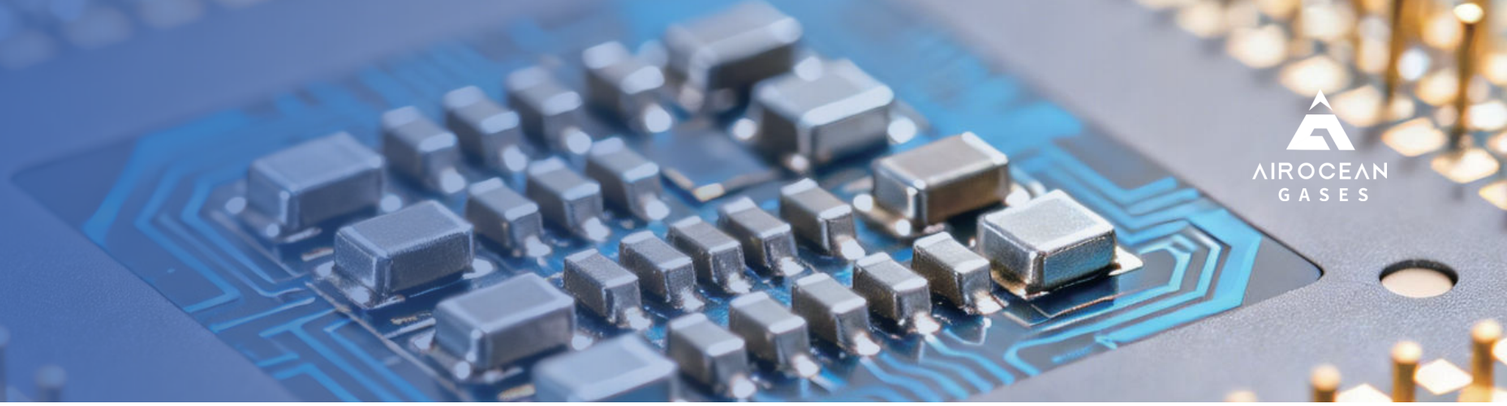
- **CYLINDER CAPACITY**
  50L high-pressure cylinder — optimized for continuous industrial and laboratory use
- **VALVE STANDARD**
  CGA 580 — the U.S. industry standard for non-flammable, non-oxidizing gases, ensuring compatibility with your existing regulators and equipment
- **FULL TRACEABILITY**
  CoA and purity certification included with every cylinder





The Etching and Dielectric Duo

SF6 and CF4



Sulfur Hexafluoride (SF_6) and Carbon Tetrafluoride (CF_4), also known as Halocarbon 14, are fluorinated gases indispensable in electrical and semiconductor manufacturing. SF6 is an exceptional dielectric

AIROCEAN GASES:YOUR PARTNER IN FLUOROCARBON SUPPLY

For both SF6 and CF4, Airocean provides full material traceability and compliance documentation. Our logistics network ensures timely and compliant international shipment, handling the unique regulatory requirements associated with these critical compounds.

SULFUR HEXAFLUORIDE (SF_6)

SF6 is non-flammable, highly stable, and possesses unparalleled insulating strength. It is 2.5 times denser than air.

Purity	99.999%	**Cylinder**	High-Pressure Steel Cylinder

DIELECTRIC INSULATION

Primary use in high-voltage circuit breakers and switchgear to prevent arcs and short circuits.

MEDICAL APPLICATIONS

Used in ophthalmology (tamponade agent) and ultrasound contrast agents.

MAGNESIUM CASTING

Used as an inert gas to prevent oxidation during the casting of magnesium parts.







CARBON TETRAFLUORIDE (CF_4)

CF4 a colorless, odorless gas, is the most stable perfluorocarbon. It is vital for plasma etching in microelectronics.

Purity	99.999%	**Cylinder**	High-Pressure Steel Cylinder

PLASMA ETCHING

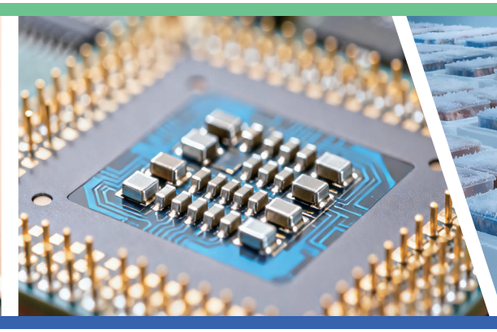
Used extensively in the semiconductor industry to etch silicon and silicon dioxide layers in chip fabrication.

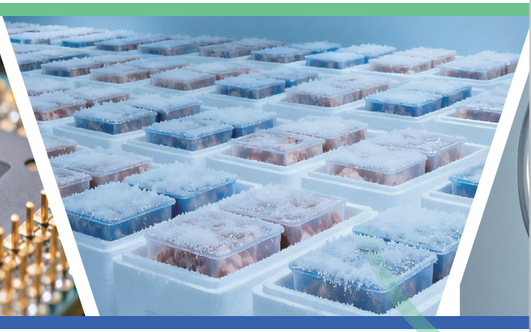
MEDICAL APPLICATIONS

Used in mixtures for cleaning CVD (Chemical Vapor Deposition) chambers.

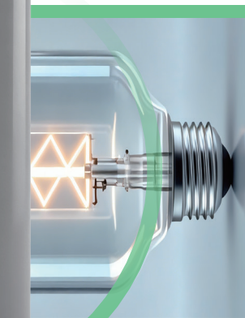
MAGNESIUM CASTING

Component in specialized low- temperature refrigerant blends.









The Invisible Layer Behind Every Modern Chip

Tungsten Hexafluoride (WF6)

Tungsten Hexafluoride is the essential precursor gas for chemical vapor deposition (CVD) of tungsten metal in semiconductor manufacturing. As device geometries shrink below 10nm, the precision of WF6 delivery and purity directly determines yield, reliability, and performance of logic and memory chips.

CORE APPLICATIONS

CVD TUNGSTEN PLUG FILL

WF6 is the industry-standard precursor for filling tungsten contacts and vias in multilayer interconnect structures — the microscopic wiring inside every modern processor and memory chip.

3D NAND & DRAM MANUFACTURING

Critical for high-aspect-ratio tungsten deposition in 3D NAND flash stacks and DRAM cell structures, enabling higher memory density and performance.

METAL GATE ELECTRODES

Used in gate-last processes for high-k/metal gate transistors at leading-edge nodes (7nm, 5nm, 3nm).

R&D & PROCESS DEVELOPMENT

Supports next-generation deposition research at semiconductor equipment companies and university cleanrooms.



PERFORMANCE CHARACTERISTICS

WF_6

AIROCEAN'S WF_6

Semiconductor-grade tungsten hexafluoride with ultra-low moisture and metallic impurity specifications for fab-ready delivery.

Chemical Formula	WF_6	**Purity**	99.999%

U.S. LOCAL WAREHOUSE & LOGISTICS

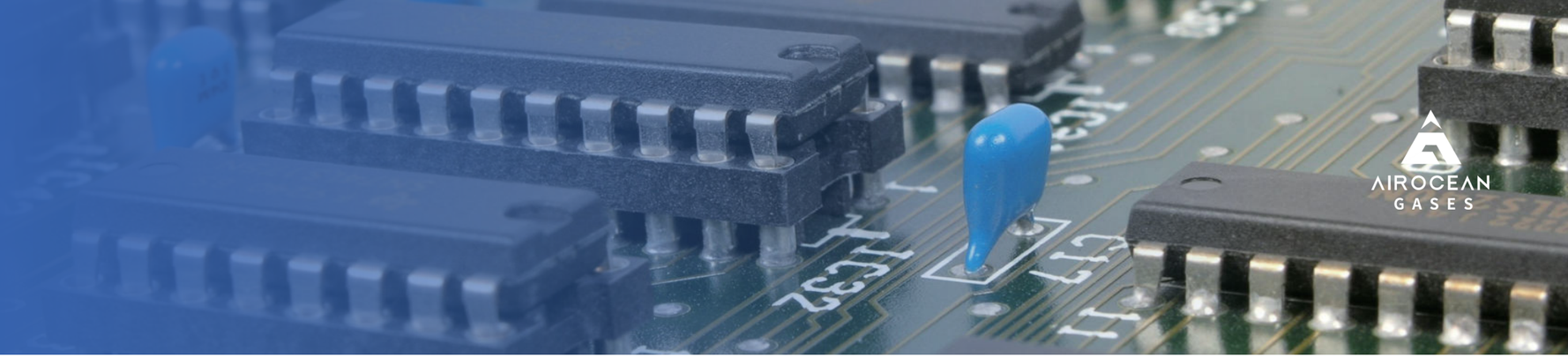
SEMICONDUCTOR FABS CAN'T AFFORD SUPPLY DELAYS. AIROCEAN'S U.S.-BASED INVENTORY AND LOGISTICS INFRASTRUCTURE ENSURES WF6 IS ALWAYS READY WHEN YOUR PROCESS DEMANDS IT.

- **INVERNTORY READY TO SHIP**
  On-hand WF6 stock at our U.S. warehouse — no overseas lead times
- **DIRECT FROM SOURCE**
  Long-term partnerships with production facilities ensure consistent purity and priority allocation
- **FULL COMPLIANCE DOCUMENTATION**
  CoA, SDS, and traceability records included with every shipment



Other products

Nitrous Oxide(N_2O)



APPLICATIONS

- FOOD & BEVERAGE
- MEDICAL & INDUSTRAIL
- SEMICONDUCTOR PROCESSES

QUALITY

- PURITY CERTIFICATION AVAILABLE
- TRACEABILITY FOR EACH CYLINDER

BASIC PARAMETERS (N_2O)

N_2O

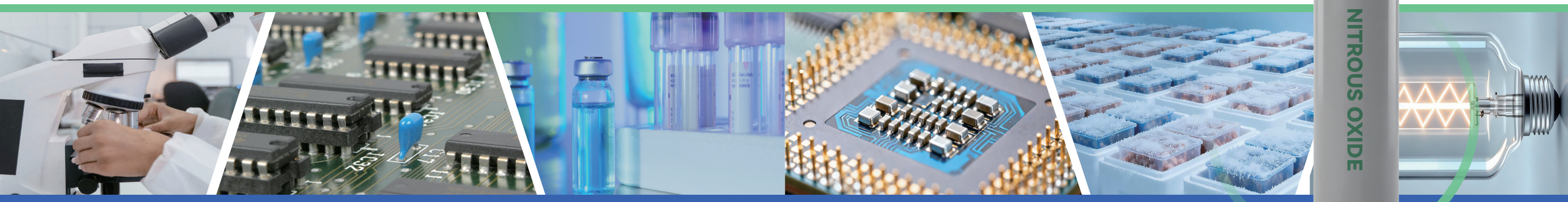
AIROCEAN'S N_2O

Available in various packaging options including cylinders and bulk to ensure uninterrupted supply.

Purity	99.5%–99.9995%	**Package**	Cylinder/Tank

WHY CHOOSE AIROCEAS

We provide comprehensive N_2O solutions across various purity levels, spanning from high-volume industrial and specialized food grade (for cream chargers) to ultra-high purity electronic grade. Backed by flexible packaging options (cylinders, tanks, and bulk) and a reliable global supply chain, we deliver consistent quality tailored to your specific industry standards.



Quality & Certification

QUALITY ASSURANCE

CERTIFIED VPROVIDED FOR ALL CYLINDERS IMPURITY ANALYSIS, AND TRACEABILITY

- Verified purity level
- Full batch traceability
- Multi-parameter impurity measurement
- Documented filling and inspection
- Standardized QA procedures

ANALYTICAL METHODS ▶

- GC/MS
- Moisture analysis
- RGA / Mass spectrometry
- Hydrocarbon detection



U.S. LOCAL WAREHOUSE

- Pennsylvania-based facility
- Regular stock of all product lines
- Hazard-compliant storage

LOGISTICS & DELIVERY

- Nationwide shipment options
- Flexible MOQs
- Fast scheduling & customer support

VALVE STANDARDS

BY GAS TYPE

CGA 580 | CGA 350 | CGA 590 | CGA 716

AIROCEAN GASES

GAS PRODUCT LINES

8 specialty

PURITY UP TO

99. 9999%

KEY FACTS

- Flexible Sourcing for Additional Gas Products
- Purity up to 99.9999%
- Strict Quality Control
- U.S. Local Warehouse
- Reliable inventory, flexible order quantities,and short lead times
- Moisture, oxygen， hydrocarbons and trace metals monitored to ppm/ppb
- Certified COAs provided for all cylinders, impurity analysis, and traceability.

